

# Test Verification Summary

The table below shows pass (✓), fail (✗), skip (○), or not-applicable (—) for each of the 11 test categories across all cases. Results are produced by `test_milp.py` calling `baselines/milp.py`.

| Case                                                 | Budget alloc. | Progress | EVM signals | Term. counter | MS cert. | Payment id. | Cash bal. | $Z^*$ | Outcome | $P$ monotone | Advance |
|------------------------------------------------------|---------------|----------|-------------|---------------|----------|-------------|-----------|-------|---------|--------------|---------|
| <i>Group 1: Progress and Milestone Certification</i> |               |          |             |               |          |             |           |       |         |              |         |
| SP-1                                                 | —             | —        | —           | —             | —        | —           | —         | —     | —       | —            | —       |
| SP-3                                                 | —             | —        | —           | —             | —        | —           | —         | —     | —       | —            | —       |
| SP-J                                                 | —             | —        | —           | —             | —        | —           | —         | —     | —       | —            | —       |
| <i>Group 2: Discounting and Timing Incentives</i>    |               |          |             |               |          |             |           |       |         |              |         |
| SP-4                                                 | —             | —        | —           | —             | —        | —           | —         | —     | —       | —            | —       |
| MP-K                                                 | —             | —        | —           | —             | —        | —           | —         | —     | —       | —            | —       |
| MP-Kp                                                | —             | —        | —           | —             | —        | —           | —         | —     | —       | —            | —       |
| SP-L                                                 | —             | —        | —           | —             | —        | —           | —         | —     | —       | —            | —       |
| SP-Lp                                                | —             | —        | —           | —             | —        | —           | —         | —     | —       | —            | —       |
| SP-M                                                 | —             | —        | —           | —             | —        | —           | —         | —     | —       | —            | —       |
| SP-Mp                                                | —             | —        | —           | —             | —        | —           | —         | —     | —       | —            | —       |
| <i>Group 3: Advance Payments and Recovery</i>        |               |          |             |               |          |             |           |       |         |              |         |
| SP-2                                                 | —             | —        | —           | —             | —        | —           | —         | —     | —       | —            | —       |
| SP-2a                                                | —             | —        | —           | —             | —        | —           | —         | —     | —       | —            | —       |
| SP-2b                                                | —             | —        | —           | —             | —        | —           | —         | —     | —       | —            | —       |
| SP-2c                                                | —             | —        | —           | —             | —        | —           | —         | —     | —       | —            | —       |
| SP-2d                                                | —             | —        | —           | —             | —        | —           | —         | —     | —       | —            | —       |
| SP-A                                                 | —             | —        | —           | —             | —        | —           | —         | —     | —       | —            | —       |
| SP-B                                                 | —             | —        | —           | —             | —        | —           | —         | —     | —       | —            | —       |
| <i>Group 4: Retention</i>                            |               |          |             |               |          |             |           |       |         |              |         |
| SP-C                                                 | —             | —        | —           | —             | —        | —           | —         | —     | —       | —            | —       |
| SP-D                                                 | —             | —        | —           | —             | —        | —           | —         | —     | —       | —            | —       |
| <i>Group 5: Counter-Trajectory Cases (Active)</i>    |               |          |             |               |          |             |           |       |         |              |         |
| SP-E                                                 | —             | —        | —           | —             | —        | —           | —         | ○     | —       | —            | —       |
| SP-F                                                 | —             | —        | —           | —             | —        | —           | —         | ○     | —       | —            | —       |
| SP-G                                                 | —             | —        | —           | —             | —        | —           | —         | ○     | —       | —            | —       |
| <i>Group 6: Forced and Starvation Termination</i>    |               |          |             |               |          |             |           |       |         |              |         |
| SP-5                                                 | —             | —        | —           | —             | —        | —           | —         | —     | —       | —            | —       |
| SP-H                                                 | —             | —        | —           | —             | —        | —           | —         | —     | —       | —            | —       |
| SP-I                                                 | —             | —        | —           | —             | —        | —           | —         | ○     | —       | —            | —       |
| <i>Group 7: Multi-Project Mixed Outcomes</i>         |               |          |             |               |          |             |           |       |         |              |         |
| MP-3                                                 | —             | —        | —           | —             | —        | —           | —         | —     | —       | —            | —       |
| MP-A                                                 | —             | —        | —           | —             | —        | —           | —         | —     | —       | —            | —       |
| MP-B                                                 | —             | —        | —           | —             | —        | —           | —         | —     | —       | —            | —       |
| <i>Group 8: Multi-Project Portfolio Optimisation</i> |               |          |             |               |          |             |           |       |         |              |         |
| MP-2                                                 | —             | —        | —           | —             | —        | —           | —         | —     | —       | —            | —       |

continued ...

| Case                                          | <i>Budget alloc.</i> | <i>Progress</i> | <i>EVM signals</i> | <i>Term. counter</i> | <i>MS cert.</i> | <i>Payment id.</i> | <i>Cash bal.</i> | <i>Z*</i> | <i>Outcome</i> | <i>P monotone</i> | <i>Advance</i> |
|-----------------------------------------------|----------------------|-----------------|--------------------|----------------------|-----------------|--------------------|------------------|-----------|----------------|-------------------|----------------|
| MP-4                                          | —                    | —               | —                  | —                    | —               | —                  | —                | —         | —              | —                 | —              |
| MP-C                                          | —                    | —               | —                  | —                    | —               | —                  | —                | —         | —              | —                 | —              |
| MP-D                                          | —                    | —               | —                  | —                    | —               | —                  | —                | —         | —              | —                 | —              |
| <i>Group 9: Staggered-Start Multi-Project</i> |                      |                 |                    |                      |                 |                    |                  |           |                |                   |                |
| MP-1                                          | —                    | —               | —                  | —                    | —               | —                  | —                | —         | —              | —                 | —              |
| MP-E                                          | —                    | —               | —                  | —                    | —               | —                  | —                | —         | —              | —                 | —              |

# Figure Legends

All figures share the following visual language.

| Panel            | Meaning                                                                                                                                                                                                                                                                                                                                                                               |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EVM (left)       |  BCWS<br> BCWP<br> ACWP<br> $\theta_j$ thresholds |
|                  | Red dashed vertical = termination period.                                                                                                                                                                                                                                                                                                                                             |
| Profile (centre) |  $x_{i,t}$ spend (down)<br> Inflows: $R^{\text{net}}$ , $A_i$ , $R^{\text{ret}}$ (up)                                                                                                                               |
|                  |                                                                                                                                                                                                                                                                                                                                                                                       |
| SPI/CPI (right)  |  = SPI (green),  = CPI (orange), red dashed = reference 1.                                                                                                                                                          |
| Portfolio left   | Cyan step-line = $B_t$ ; grey dashed = $B_1$ reference.                                                                                                                                                                                                                                                                                                                               |
| Portfolio centre | Green bars = inflow, red bars = outflow, blue line with $\circ$ = net per period.                                                                                                                                                                                                                                                                                                     |
| Portfolio right  | Blue line with $\bullet$ = cumulative discounted NCF.                                                                                                                                                                                                                                                                                                                                 |

X-axes scale to each case's horizon  $H$ . Middle project panel uses symmetric  $\pm \text{BAC}_i$  range. Portfolio panels use data-driven ranges.

## Conventions

- $\gamma = 0.95$  throughout.
- Advance  $A_i$  arrives at  $t = s_i - 1$  (period 0 when  $s_i = 1$ ).
- Payment identity:  $A_i + \sum_j R_{i,j}^{\text{net}} + R_i^{\text{ret}} = CP_i$  checked for completed projects only.
- Retention released as a lump at the period the final milestone certifies.
- Advance recovery deductions:  $\delta^{\text{rec}} \phi_j CP_i$  (pre-retention base, FIDIC convention).

# Cases

## Group 1: Progress and Milestone Certification

### SP-1 (Group G1)

|                                  |                    |
|----------------------------------|--------------------|
| $n, H$                           | 1, 4               |
| BAC, $CP$                        | 100, 120           |
| $s_i, f_i, D^{\text{plan}}$      | 1, 4, 4            |
| $\bar{\eta}$                     | 1                  |
| $M, \theta$                      | 2, (0.5, 1)        |
| $\phi, e_{i,j}$                  | (0.5, 0.5), (1, 3) |
| $\mu, \Omega, \tau^{\text{tol}}$ | 0.3, 1, 2          |
| $B_1, \gamma$                    | 300, 0.95          |
| $Z^*$                            | 19.025             |
| Outcome                          | completed          |

| $t$ | $x$ | $P$ | BCWS | BCWP | ACWP | SPI   | CPI | $\tau$ | MS | $R_{\text{net}}$ |
|-----|-----|-----|------|------|------|-------|-----|--------|----|------------------|
| 1   | 50  | 0.5 | 0.25 | 0.5  | 0.5  | 2     | 1   | 2      | 1  | 60               |
| 2   | 0   | 0.5 | 0.5  | 0.5  | 0.5  | 1     | 1   | 2      | -- | 0                |
| 3   | 50  | 1   | 0.75 | 1    | 1    | 1.333 | 1   | 2      | 2  | 60               |
| 4   | 0   | 1   | 1    | 1    | 1    | 1     | 1   | 2      | -- | 0                |

Portfolio strip.

| $t$ | $B_t$ | $\Sigma \text{In}$ | $\Sigma \text{Out}$ | $\Sigma \text{Net}$ | $\gamma^{t-1} \text{Net}$ | $Z_t^{\text{cum}}$ |
|-----|-------|--------------------|---------------------|---------------------|---------------------------|--------------------|
| 1   | 310   | 60                 | 50                  | 10                  | 10                        | 10                 |
| 2   | 310   | 0                  | 0                   | 0                   | 0                         | 10                 |
| 3   | 320   | 60                 | 50                  | 10                  | 9.025                     | 19.025             |
| 4   | 320   | 0                  | 0                   | 0                   | 0                         | 19.025             |

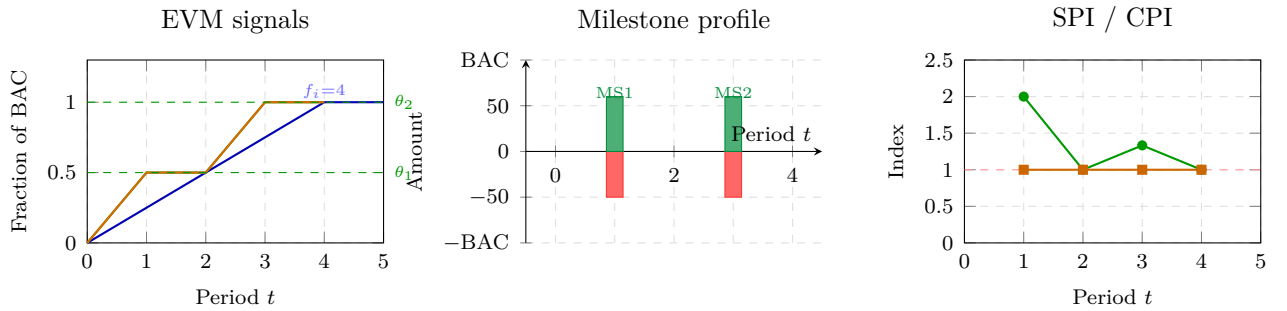


Figure 1: SP-1 project view (single-project). Outcome: **completed**.  $Z^* = 19.025$ .

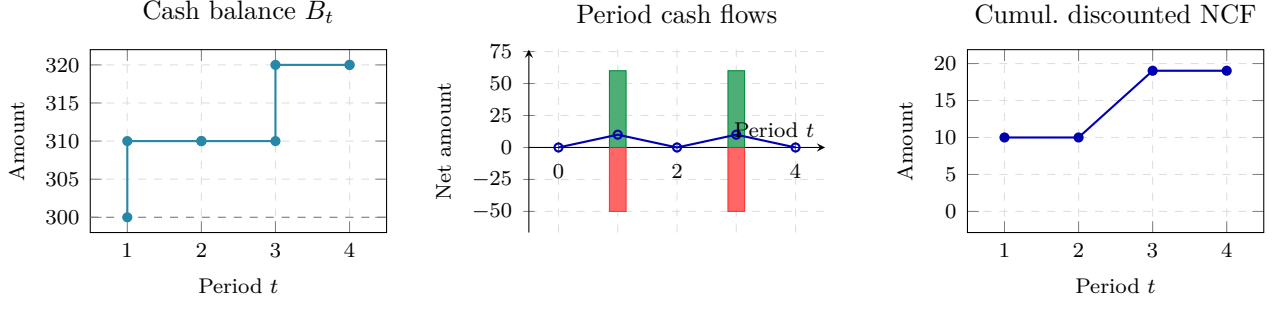


Figure 2: SP-1 portfolio view.  $Z^* = 19.025$ .

### SP-3 (Group G1)

|                                  |                                  |
|----------------------------------|----------------------------------|
| $n, H$                           | 1, 6                             |
| BAC, $CP$                        | 60, 90                           |
| $s_i, f_i, D^{\text{plan}}$      | 1, 6, 6                          |
| $\bar{\eta}$                     | 1                                |
| $M, \theta$                      | 3, (0.333, 0.667, 1)             |
| $\phi, e_{i,j}$                  | (0.333, 0.333, 0.333), (1, 3, 5) |
| $\mu, \Omega, \tau^{\text{tol}}$ | 0.5, 2, 2                        |
| $B_1, \gamma$                    | 300, 0.95                        |
| $Z^*$                            | 27.17                            |
| Outcome                          | completed                        |

| $t$ | $x$ | $P$   | BCWS  | BCWP  | ACWP  | SPI   | CPI | $\tau$ | MS | $R_{\text{net}}$ |
|-----|-----|-------|-------|-------|-------|-------|-----|--------|----|------------------|
| 1   | 20  | 0.333 | 0.167 | 0.333 | 0.333 | 2     | 1   | 2      | 1  | 30               |
| 2   | 0   | 0.333 | 0.333 | 0.333 | 0.333 | 1     | 1   | 2      | -- | 0                |
| 3   | 20  | 0.667 | 0.5   | 0.667 | 0.667 | 1.333 | 1   | 2      | 2  | 30               |
| 4   | 0   | 0.667 | 0.667 | 0.667 | 0.667 | 1     | 1   | 2      | -- | 0                |
| 5   | 20  | 1     | 0.833 | 1     | 1     | 1.2   | 1   | 2      | 3  | 30               |
| 6   | 0   | 1     | 1     | 1     | 1     | 1     | 1   | 2      | -- | 0                |

### Portfolio strip.

| $t$ | $B_t$ | $\Sigma \text{In}$ | $\Sigma \text{Out}$ | $\Sigma \text{Net}$ | $\gamma^{t-1} \text{Net}$ | $Z_t^{\text{cum}}$ |
|-----|-------|--------------------|---------------------|---------------------|---------------------------|--------------------|
| 1   | 310   | 30                 | 20                  | 10                  | 10                        | 10                 |
| 2   | 310   | 0                  | 0                   | 0                   | 0                         | 10                 |
| 3   | 320   | 30                 | 20                  | 10                  | 9.025                     | 19.025             |
| 4   | 320   | 0                  | 0                   | 0                   | 0                         | 19.025             |
| 5   | 330   | 30                 | 20                  | 10                  | 8.145                     | 27.17              |
| 6   | 330   | 0                  | 0                   | 0                   | 0                         | 27.17              |

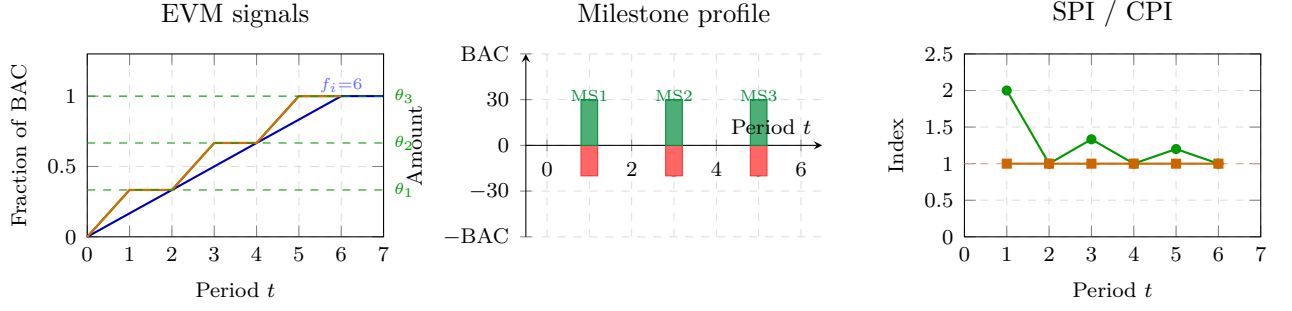


Figure 3: SP-3 project view (single-project). Outcome: **completed**.  $Z^* = 27.17$ .

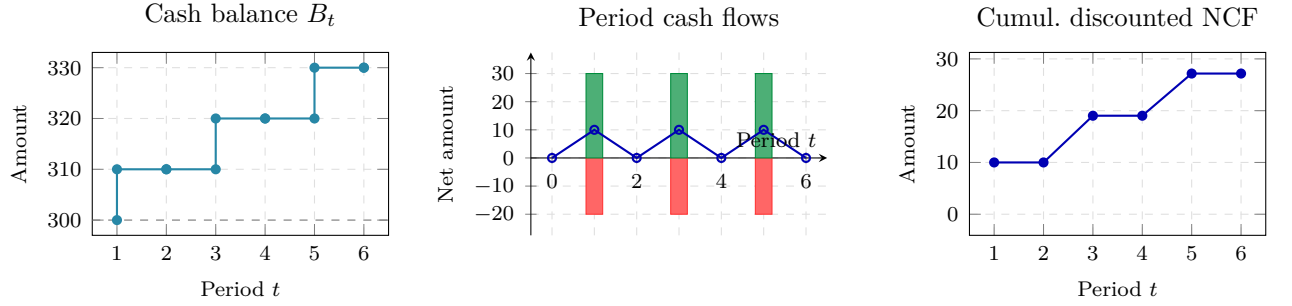


Figure 4: SP-3 portfolio view.  $Z^* = 27.17$ .

## SP-J (Group G1)

|                                  |                    |
|----------------------------------|--------------------|
| $n, H$                           | 1, 4               |
| BAC, $CP$                        | 100, 100           |
| $s_i, f_i, D^{\text{plan}}$      | 1, 4, 4            |
| $\bar{\eta}$                     | 1                  |
| $M, \theta$                      | 2, (0.5, 1)        |
| $\phi, e_{i,j}$                  | (0.5, 0.5), (1, 3) |
| $\mu, \Omega, \tau^{\text{tol}}$ | 0.5, 2, 2          |
| $B_1, \gamma$                    | 300, 0.95          |
| $Z^*$                            | 0                  |
| Outcome                          | <b>completed</b>   |

| $t$ | $x$ | $P$ | BCWS | BCWP | ACWP | SPI   | CPI | $\tau$ | MS | $R_{\text{net}}$ |
|-----|-----|-----|------|------|------|-------|-----|--------|----|------------------|
| 1   | 50  | 0.5 | 0.25 | 0.5  | 0.5  | 2     | 1   | 2      | 1  | 50               |
| 2   | 0   | 0.5 | 0.5  | 0.5  | 0.5  | 1     | 1   | 2      | -- | 0                |
| 3   | 50  | 1   | 0.75 | 1    | 1    | 1.333 | 1   | 2      | 2  | 50               |
| 4   | 0   | 1   | 1    | 1    | 1    | 1     | 1   | 2      | -- | 0                |

**Portfolio strip.**

| $t$ | $B_t$ | $\Sigma\text{In}$ | $\Sigma\text{Out}$ | $\Sigma\text{Net}$ | $\gamma^{t-1}\text{Net}$ | $Z_t^{\text{cum}}$ |
|-----|-------|-------------------|--------------------|--------------------|--------------------------|--------------------|
| 1   | 300   | 50                | 50                 | 0                  | 0                        | 0                  |
| 2   | 300   | 0                 | 0                  | 0                  | 0                        | 0                  |
| 3   | 300   | 50                | 50                 | 0                  | 0                        | 0                  |
| 4   | 300   | 0                 | 0                  | 0                  | 0                        | 0                  |

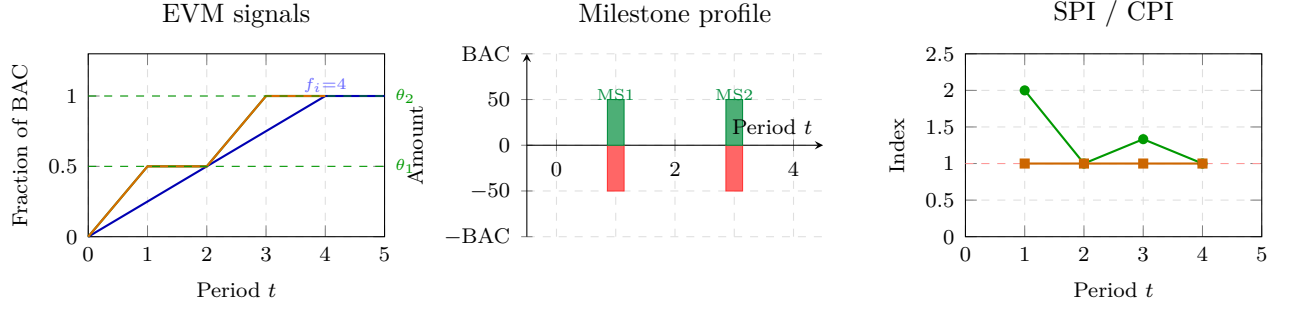


Figure 5: SP-J project view (single-project). Outcome: completed.  $Z^* = 0$ .

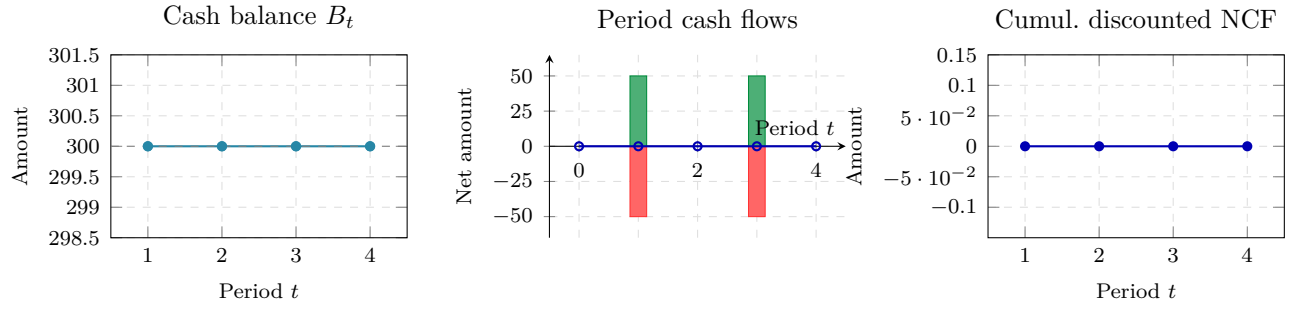


Figure 6: SP-J portfolio view.  $Z^* = 0$ .

## Group 2: Discounting and Timing Incentives

### SP-4 (Group G2)

|                                  |                    |
|----------------------------------|--------------------|
| $n, H$                           | 1, 8               |
| BAC, $CP$                        | 100, 140           |
| $s_i, f_i, D^{\text{plan}}$      | 1, 8, 8            |
| $\bar{\eta}$                     | 1                  |
| $M, \theta$                      | 2, (0.5, 1)        |
| $\phi, e_{i,j}$                  | (0.5, 0.5), (1, 5) |
| $\mu, \Omega, \tau^{\text{tol}}$ | 0.3, 1, 2          |
| $B_1, \gamma$                    | 300, 0.95          |
| $Z^*$                            | 36.29              |
| Outcome                          | completed          |

| $t$ | $x$ | $P$ | BCWS  | BCWP | ACWP | SPI   | CPI | $\tau$ | MS | $R_{\text{net}}$ |
|-----|-----|-----|-------|------|------|-------|-----|--------|----|------------------|
| 1   | 50  | 0.5 | 0.125 | 0.5  | 0.5  | 4     | 1   | 2      | 1  | 70               |
| 2   | 0   | 0.5 | 0.25  | 0.5  | 0.5  | 2     | 1   | 2      | -- | 0                |
| 3   | 0   | 0.5 | 0.375 | 0.5  | 0.5  | 1.333 | 1   | 2      | -- | 0                |
| 4   | 0   | 0.5 | 0.5   | 0.5  | 0.5  | 1     | 1   | 2      | -- | 0                |
| 5   | 50  | 1   | 0.625 | 1    | 1    | 1.6   | 1   | 2      | 2  | 70               |
| 6   | 0   | 1   | 0.75  | 1    | 1    | 1.333 | 1   | 2      | -- | 0                |
| 7   | 0   | 1   | 0.875 | 1    | 1    | 1.143 | 1   | 2      | -- | 0                |
| 8   | 0   | 1   | 1     | 1    | 1    | 1     | 1   | 2      | -- | 0                |

### Portfolio strip.

| $t$ | $B_t$ | $\Sigma \text{In}$ | $\Sigma \text{Out}$ | $\Sigma \text{Net}$ | $\gamma^{t-1} \text{Net}$ | $Z_t^{\text{cum}}$ |
|-----|-------|--------------------|---------------------|---------------------|---------------------------|--------------------|
| 1   | 320   | 70                 | 50                  | 20                  | 20                        | 20                 |
| 2   | 320   | 0                  | 0                   | 0                   | 0                         | 20                 |
| 3   | 320   | 0                  | 0                   | 0                   | 0                         | 20                 |
| 4   | 320   | 0                  | 0                   | 0                   | 0                         | 20                 |
| 5   | 340   | 70                 | 50                  | 20                  | 16.29                     | 36.29              |
| 6   | 340   | 0                  | 0                   | 0                   | 0                         | 36.29              |
| 7   | 340   | 0                  | 0                   | 0                   | 0                         | 36.29              |
| 8   | 340   | 0                  | 0                   | 0                   | 0                         | 36.29              |

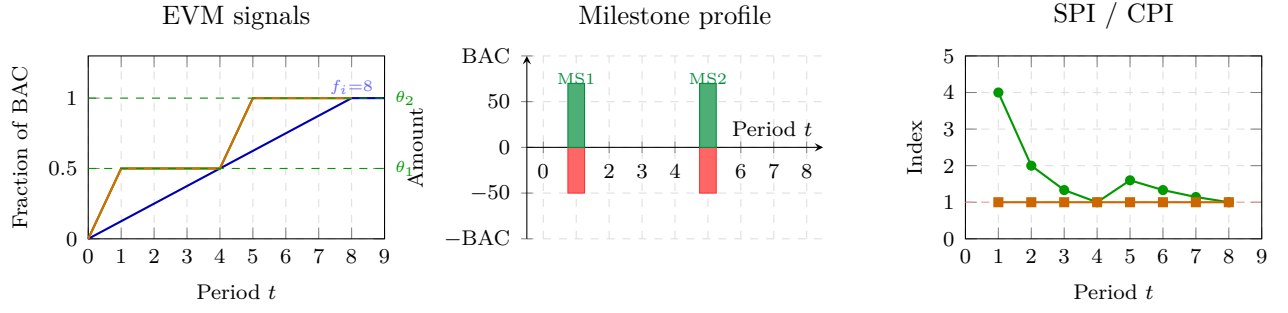


Figure 7: SP-4 project view (single-project). Outcome: **completed**.  $Z^* = 36.29$ .

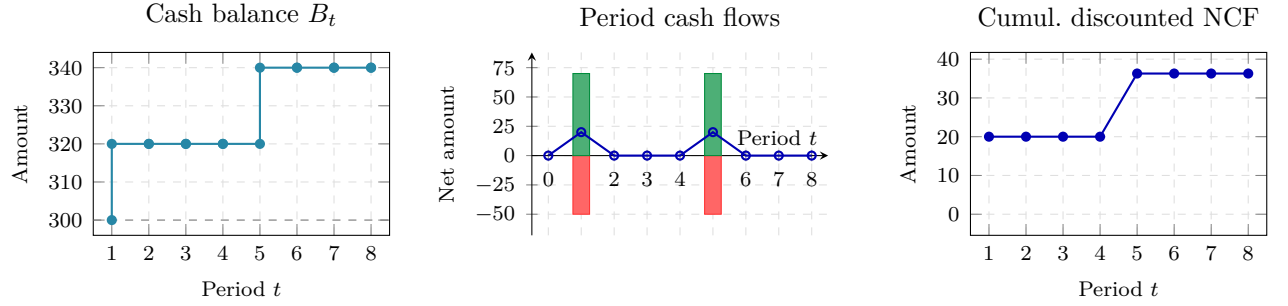


Figure 8: SP-4 portfolio view.  $Z^* = 36.29$ .

## MP-K (Group G2)

|               |                |
|---------------|----------------|
| $n, H$        | 2, 6           |
| $BAC_i$       | (60, 60)       |
| $CP_i$        | (90, 90)       |
| $B_1, \gamma$ | 300, 0.95      |
| $Z^*$         | 55.87          |
| Outcome       | both_completed |

### Project 1 timeline.

| $t$ | $x$ | $P$ | BCWS  | BCWP | ACWP | SPI | CPI | $\tau$ | MS | $R_{\text{net}}$ |
|-----|-----|-----|-------|------|------|-----|-----|--------|----|------------------|
| 1   | 30  | 0.5 | 0.167 | 0.5  | 0.5  | 3   | 1   | 2      | 1  | 63               |
| 2   | 0   | 0.5 | 0.333 | 0.5  | 0.5  | 1.5 | 1   | 2      | -- | 0                |
| 3   | 0   | 0.5 | 0.5   | 0.5  | 0.5  | 1   | 1   | 2      | -- | 0                |
| 4   | 30  | 1   | 0.667 | 1    | 1    | 1.5 | 1   | 2      | 2  | 27               |
| 5   | 0   | 1   | 0.833 | 1    | 1    | 1.2 | 1   | 2      | -- | 0                |
| 6   | 0   | 1   | 1     | 1    | 1    | 1   | 1   | 2      | -- | 0                |

### Project 2 timeline.

| $t$ | $x$ | $P$ | BCWS  | BCWP | ACWP | SPI | CPI | $\tau$ | MS | $R_{\text{net}}$ |
|-----|-----|-----|-------|------|------|-----|-----|--------|----|------------------|
| 1   | 0   | 0   | 0.167 | 0    | 0    | 0   | 1   | 2      | -- | 0                |
| 2   | 30  | 0.5 | 0.333 | 0.5  | 0.5  | 1.5 | 1   | 2      | 1  | 27               |
| 3   | 0   | 0.5 | 0.5   | 0.5  | 0.5  | 1   | 1   | 2      | -- | 0                |
| 4   | 30  | 1   | 0.667 | 1    | 1    | 1.5 | 1   | 2      | 2  | 63               |
| 5   | 0   | 1   | 0.833 | 1    | 1    | 1.2 | 1   | 2      | -- | 0                |
| 6   | 0   | 1   | 1     | 1    | 1    | 1   | 1   | 2      | -- | 0                |

Portfolio strip.

| $t$ | $B_t$ | $\Sigma \text{In}$ | $\Sigma \text{Out}$ | $\Sigma \text{Net}$ | $\gamma^{t-1} \text{Net}$ | $Z_t^{\text{cum}}$ |
|-----|-------|--------------------|---------------------|---------------------|---------------------------|--------------------|
| 1   | 333   | 63                 | 30                  | 33                  | 33                        | 33                 |
| 2   | 330   | 27                 | 30                  | -3                  | -2.85                     | 30.15              |
| 3   | 330   | 0                  | 0                   | 0                   | 0                         | 30.15              |
| 4   | 360   | 90                 | 60                  | 30                  | 25.721                    | 55.871             |
| 5   | 360   | 0                  | 0                   | 0                   | 0                         | 55.871             |
| 6   | 360   | 0                  | 0                   | 0                   | 0                         | 55.871             |

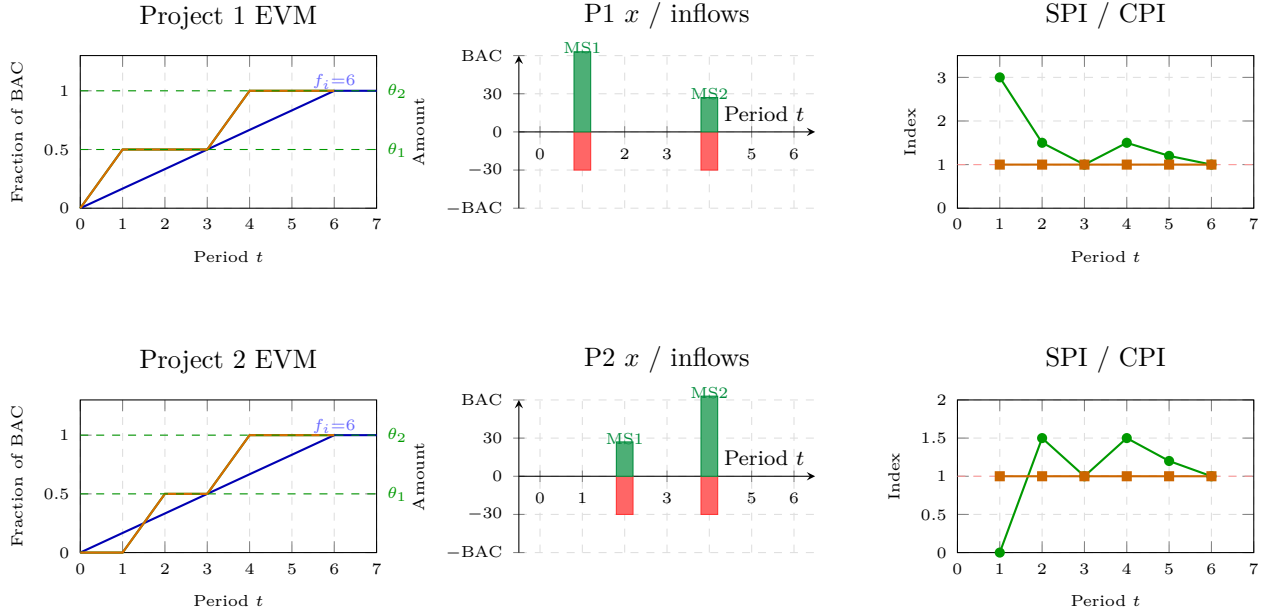


Figure 9: MP-K project view (2-project). Outcome:  $\text{both}_{\text{completed}}.Z^* = 55.87$ .